ModelIt!

Student Activity Pack

The Plastic Paradox

Modeling Microplastic Bioaccumulation, Ocean Gyres, and Decomposition Timescales

Grade 11L2

NGSS HS-ESS3-4, HS-LS2-6

Cause and Effect / Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens to a plastic bottle after it enters the ocean — does it ever truly disappear?**

2. **Why might eating seafood expose you to chemicals from plastic pollution thousands of miles away?**

3. **Draw a diagram showing how you think plastic moves from land to the ocean and through the marine food web.**

4. **If we stopped producing plastic today, would the ocean plastic problem go away? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Plastic Production and Disposal Rate
- UV and Mechanical Fragmentation
- Ocean Gyre Concentration
- Microplastic Ingestion Rate
- Trophic-Level Toxin Concentration
- Marine Ecosystem Health

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trajectory

11 million tons/year entering oceans, gyre concentration increasing

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Source Reduction

Plastic input cut 50%, no active cleanup

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cleanup + Reduction

50% input reduction plus 10,000 tons/year removed from gyres

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Nanoplastic Formation — As microplastics continue to fragment below 1 micrometer, they become nanoplastics capable of crossing cell membranes and the blood-brain barrier — a contamination pathway that is poorly understood but potentially far more dangerous than microplastics

Chemical Leaching Rate — Plastic additives including plasticizers, flame retardants, and UV stabilizers leach into surrounding water and organisms over time, creating a chemical exposure pathway independent of physical ingestion

Sediment Burial Rate — Microplastics that sink to the ocean floor become buried in sediments, creating a long-term contamination reservoir that can be re-released by storms, trawling, and bioturbation — the deep ocean is becoming a permanent plastic sink

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does a material designed to last forever become the greatest threat to the ecosystems that can't break it down? Use evidence from your simulation to support your answers.

1. **Why does your model show that ocean cleanup alone cannot solve the microplastic crisis, even with advanced technology?**

2. **How does the concept of biomagnification reveal that low-concentration pollution at the base of the food web becomes a high-concentration crisis for top predators?**

3. **What feedback loops in your model make the microplastic problem self-reinforcing once a certain contamination threshold is crossed?**

4. **How does the mismatch between plastic's decomposition timescale (centuries) and human decision-making timescale (years) create a unique environmental challenge?**

5. **What does your model suggest about the environmental justice dimensions of plastic pollution — who produces it, who suffers from it, and who has the power to change it?**

8 STEM Challenge: Design a Microplastic Contamination Reduction System

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YOUR ENGINEERING MISSION

Design an integrated strategy that combines source reduction, material innovation, and targeted cleanup to reduce microplastic contamination in a coastal marine ecosystem by 80% within 20 years.

THE SCENARIO

A major coastal fishery has detected rising microplastic levels in commercially important species, threatening both ecosystem health and the $500 million regional fishing industry. Your team has been hired to design a comprehensive microplastic reduction plan that addresses the problem from production to ocean cleanup.

Engineering Design Thinking

1. WHAT DATA WOULD YOU NEED TO IDENTIFY THE PRIMARY SOURCES AND PATHWAYS OF PLASTIC ENTERING THIS COASTAL ECOSYSTEM?

2. HOW WOULD YOU PRIORITIZE INTERVENTION STRATEGIES BASED ON COST-EFFECTIVENESS, IMPLEMENTATION TIMELINE, AND IMPACT MAGNITUDE?

3. WHAT CLEANUP TECHNOLOGIES COULD TARGET MICROPLASTICS IN COASTAL WATERS WITHOUT HARMING MARINE ORGANISMS?

4. HOW WOULD YOU MEASURE AND VERIFY THE SUCCESS OF YOUR REDUCTION STRATEGY ACROSS A 20-YEAR TIMELINE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Marine Pollution Scientists study the sources, transport, fate, and ecological effects of contaminants in ocean environments, using field sampling, laboratory analysis, and computational modeling to inform policy. They work for agencies like NOAA, EPA, and the UN Environment Programme, earning $70,000–$130,000/year. Materials Scientists developing biodegradable plastic alternatives at companies and research labs earn $75,000–$150,000/year.